

YEAR 7 MATHEMATICS · CHAPTER 10

Analysing data

10.01 Interpreting graphs**10.03** Dot plots**10.05** The mean and mode**10.07** Analysing dot plots and stem-and-leaf plots**10.02** Misleading graphs**10.04** Stem-and-leaf plots**10.06** The median and range**10.08** Comparing data sets

Name _____

Class _____

Teacher _____

How to use this booklet

- Each lesson starts with **learning intentions** and **success criteria**. Write the **key terms** as your teacher explains them, and use the **notes** box.
- Work through the levels in order: Level 1, then Level 2, then Level 3. Extension questions stretch you further.
- Some questions ask you to **draw** a graph, dot plot or stem-and-leaf plot. Use the grid or template given and a ruler.
- At the end of each lesson, **show off your skill**: choose one question, then tear off the ticket and hand it in.
- Missed a lesson? Each lesson ends with a **summary** that has a worked example for every level. Start there, then try the questions.

WE DO

Grey boxes with a dark border. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box and write your answer.

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In this chapter you will

- interpret a variety of statistical graphs, including divided bar graphs, sector graphs and line graphs
- identify graphs that are misleading
- draw stem-and-leaf plots and dot plots
- calculate the mean, mode, median and range of a set of data, including data shown on graphs and plots
- compare data sets using statistics, including identifying outliers
- use primary and secondary data

NSW Mathematics K–10 Syllabus (2022), Stage 4 Statistics: MA4-DAT-C-01 classifies and displays data using a variety of graphical representations; MA4-DAT-C-02 analyses simple datasets using measures of centre, range and shape of the data.



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.01

Interpreting graphs

Date _____

Learning intentions

We are learning to:

- read and interpret picture, column, sector, line and divided bar graphs
- explain what a graph tells us about real data

Success criteria

I can:

- ☐ use a **key** and a **scale** to read values from a graph
- ☐ estimate amounts from a **sector graph** using fractions of a circle
- ☐ read percentages from a **divided bar graph** and give reasons for patterns

Key terms Write the definition as your teacher explains it.

Key

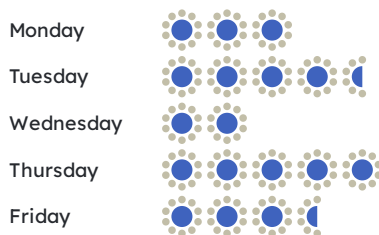
Sector graph

Divided bar graph

Notes

Level 1 · Getting started

Books borrowed from the library



 = 4 books

WE DO

With your teacher: copy what they write on the board.

Example 1

How many books were borrowed on Tuesday?

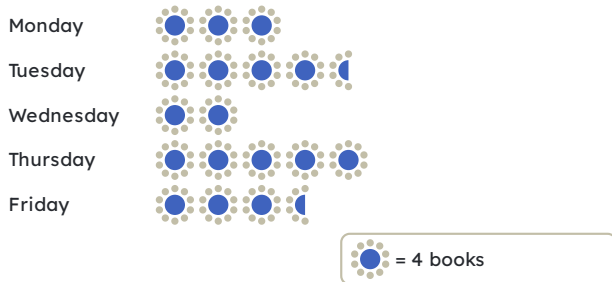
Example 2

How many more books on Thursday than Wednesday?

YOU DO

On your own: Set A. Use the library picture graph.

Books borrowed from the library

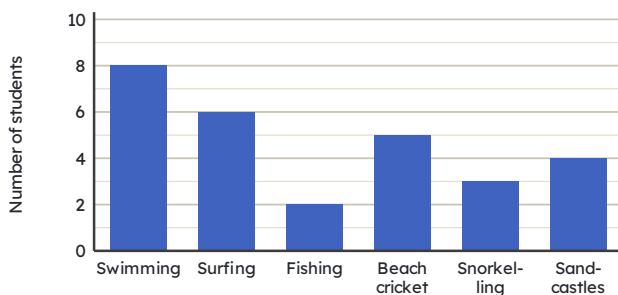


- 1 What does 1 symbol stand for?
- 2 What does $\frac{1}{2}$ a symbol stand for?
- 3 How many books on Monday?
- 4 How many books on Wednesday?
- 5 How many books on Friday?
- 6 Which day had the most books?
- 7 Which day had the fewest books?
- 8 How many more on Friday than Monday?
- 9 How many on Monday and Tuesday?
- 10 How many symbols show 16 books?

YOU DO

On your own: Set B. Use the column graph.

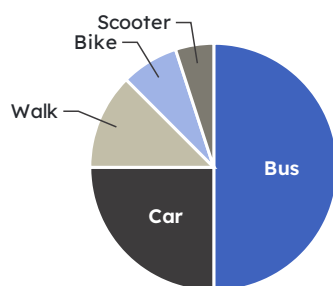
Favourite beach activity of 7K



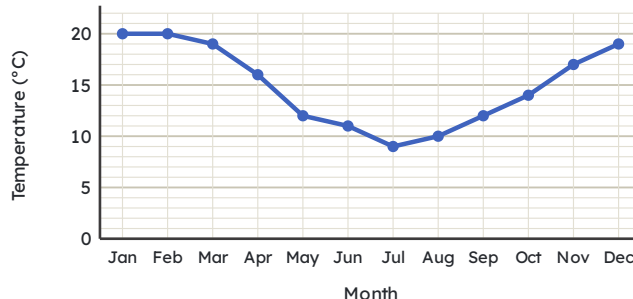
- 1 How many chose surfing?
- 2 How many chose fishing?
- 3 Which was the most popular?
- 4 Which was the least popular?
- 5 Which activity had 5 votes?
- 6 How many chose snorkelling?
- 7 How many more swam than fished?
- 8 How many chose surfing or fishing?
- 9 How many students are in 7K?
- 10 What does 1 gridline stand for?

Level 2 · Building up

How 80 Year 7 students travel to school



Mean minimum temperature, Kingscliff



WE DO With your teacher: copy what they write on the board.

Example 3 What fraction of the students catch the bus? How many students is that?

Example 4 Which month is coldest? How much colder is June than March?

YOU DO On your own: Set C. Questions 1–4 use the sector graph above.

1 Which way of travel is the most common?

3 Estimate how many students walk.

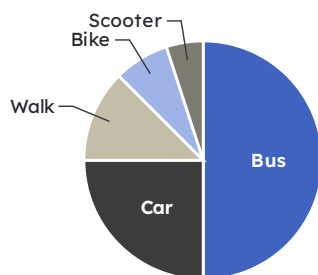
2 Estimate how many students travel by car.

4 True or false: fewer than 10 ride a bike.

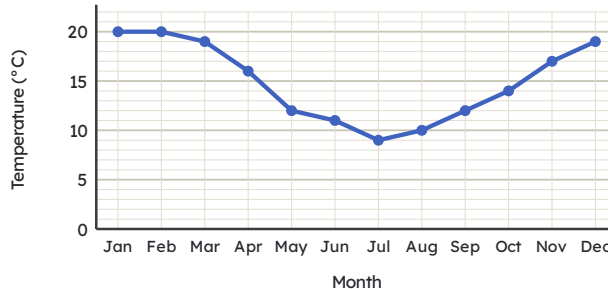
YOU DO

On your own: Set C continued. Question 5 uses the sector graph and 6–10 use the line graph.

How 80 Year 7 students travel to school



Mean minimum temperature, Kingscliff



- 5 How many students do **not** catch the bus?

- 8 Which month has a mean minimum of 14 °C?

- 6 Which 2 months are warmest? What is their mean minimum?

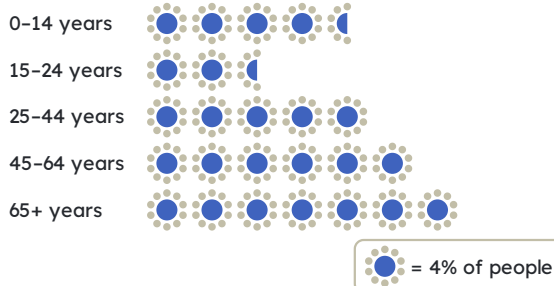
- 9 Which 2 months in a row have the biggest rise?

- 7 Name 2 months with the same mean minimum.

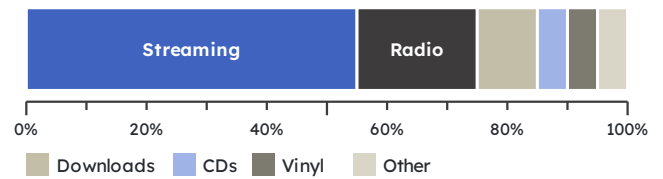
- 10 Which 2 months in a row have the biggest drop?

Level 3 · Challenge

Ages of people in a coastal town



How Australians listen to music



WE DO With your teacher: copy what they write on the board.

Example 5 What percentage of the town is 45+?
How many is that out of 5000?

Example 6 What % stream music? How might this look different 20 years ago?

YOU DO On your own: Set D. Use the picture graph above.

1 What percentage of people are aged 65+?

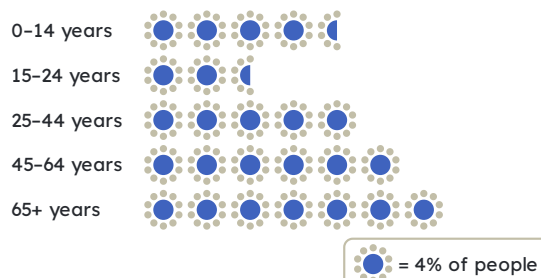
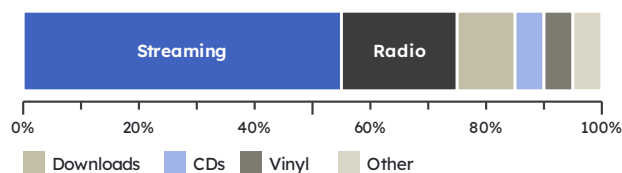
3 Town of 5000 people: how many are 0-14?

2 What percentage of people are under 25?

4 How many symbols would show 30% of the town?

YOU DO

On your own: Set D continued. Question 5 uses the picture graph and 6–10 use the divided bar graph.

Ages of people in a coastal town

How Australians listen to music


5 Why might 65+ be the biggest group in this town?

8 Survey of 400: how many use downloads?

6 Which 2 ways of listening add up to 75% of people?

9 Of 400: how many more stream than radio?

7 What fraction of people listen to the radio?

10 How might this graph change in 10 years? Why?

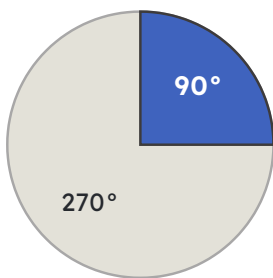
Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

In a survey of 72 students, the **Soccer** sector of a sector graph has an angle of 90° . How many students chose soccer?

1. A full circle is 360° , so 90° is $90/360 = \frac{1}{4}$ of the circle
2. $\frac{1}{4}$ of 72 = $72 \div 4 = 18$



90° out of 360°
= $\frac{1}{4}$ of the circle
= $\frac{1}{4}$ of the students

Answer: 18 students chose soccer

YOU DO

On your own: read the worked example, then try these.

- E1** A survey of 240 people has a **Tennis** sector of 60° . How many chose tennis?

- E2** In a survey of 180 people, 45 chose pizza. What angle should the pizza sector be?

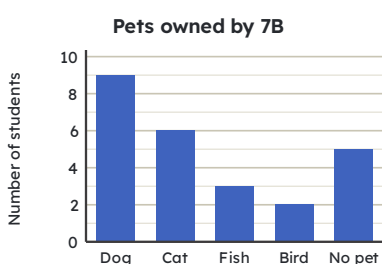
- E3** One symbol = 8 students. How many students do $3\frac{3}{4}$ symbols show? How would you show 20 students?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

How many students in 7B own a cat?

Level 2

What fraction of the class owns a dog?

Level 3

What % of 7B has no pet? How many of 200 Year 7s would you expect to have no pet?

Lesson summary

Missed the lesson? Start here.

Reading any graph

1. Read the **title**: what is the graph about?
2. Check the **key** or the **scale**.
3. Use a ruler to line up bars and points with the scale.
4. Answer with the right **units** (students, °C, %).

Level 1

In a picture graph,  = 10 students. A row shows 3½ symbols. How many students is that?

1. 3 full symbols = $3 \times 10 = 30$
2. Half a symbol = $10 \div 2 = 5$
3. $30 + 5 = 35$

Row

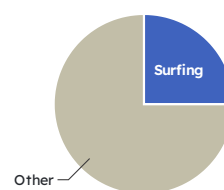

 = 10 students

Answer: 35 students

Level 2

A sector graph shows 80 people. The **Surfing** sector is a quarter of the circle. How many chose surfing?

1. A quarter means divide by 4
2. $80 \div 4 = 20$

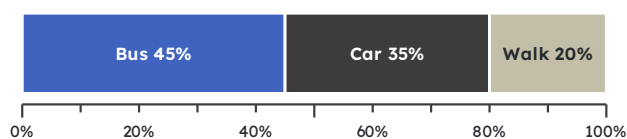


Answer: 20 people

Level 3

A divided bar graph shows 45% of 400 students catch the bus. How many students is that?

1. 45% means 45 out of every 100
2. 400 has 4 hundreds, so $45 \times 4 = 180$

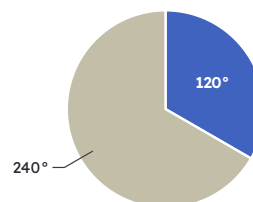


Answer: 180 students

Extension

A 120° sector shows the results for 90 people. How many people is that?

1. $120/360 = \frac{1}{3}$ of the circle
2. $\frac{1}{3}$ of 90 = $90 \div 3 = 30$



Answer: 30 people

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LESSON
10.02

Misleading graphs

Date _____

Learning intentions

We are learning to:

- spot features that make a graph misleading
- explain the false impression a misleading graph gives
- redraw a misleading graph so it is fair

Success criteria

I can:

- ☐ check whether the vertical axis starts at 0 and has an even scale
- ☐ explain how a graph makes a difference look bigger or smaller than it really is
- ☐ redraw a graph with a fair scale, title and labels

Key terms Write the definition as your teacher explains it.

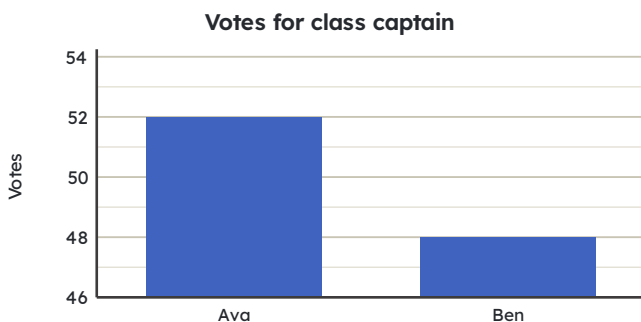
Misleading graph

Scale

Axis

Notes

Level 1 · Getting started


WE DO

With your teacher: copy what they write on the board.

Example 1

What is misleading about this graph?

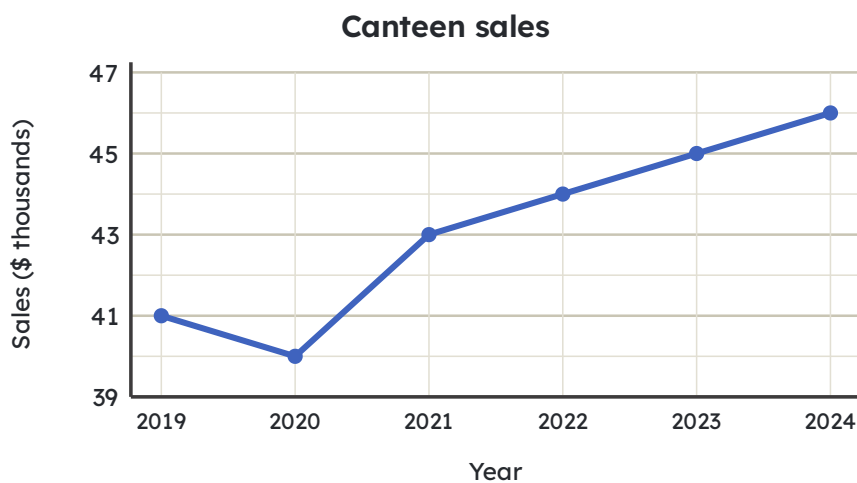
Example 2

Ava's bar looks 3 times as tall as Ben's. How many more votes did Ava really get?

YOU DO On your own: Set A. Is each feature fair or misleading? Write F or M.

- | | | | |
|--|----------------------|---|----------------------|
| 1 The vertical axis starts at 0. | <input type="text"/> | 6 The graph has a clear title. | <input type="text"/> |
| 2 The vertical axis starts at 90, not 0. | <input type="text"/> | 7 Some years are left out of a line graph. | <input type="text"/> |
| 3 The scale goes 0, 10, 20, 50, 100 with equal gaps. | <input type="text"/> | 8 A tilted 3D pie makes the front sector look bigger. | <input type="text"/> |
| 4 Both axes are labelled with units. | <input type="text"/> | 9 The gridlines are evenly spaced. | <input type="text"/> |
| 5 One bar is drawn wider than the others. | <input type="text"/> | 10 A picture is drawn taller and wider to show a bigger value. | <input type="text"/> |

YOU DO On your own: Set B. Use the canteen sales line graph.

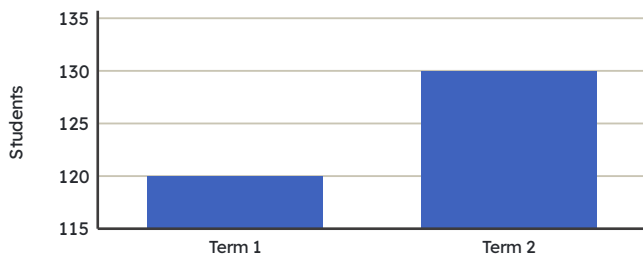


- | | | | |
|--|----------------------|---|----------------------|
| 1 What were sales in 2019? | <input type="text"/> | 6 On the graph, 2024 looks how many times higher than 2020? | <input type="text"/> |
| 2 In which year were sales lowest? | <input type="text"/> | 7 True or false: sales more than doubled. | <input type="text"/> |
| 3 What does the vertical axis start at? | <input type="text"/> | 8 Which year had the biggest rise? | <input type="text"/> |
| 4 Should it start at 0? (yes/no) | <input type="text"/> | 9 Are the axes labelled? (yes/no) | <input type="text"/> |
| 5 How much did sales rise from 2020 to 2024? | <input type="text"/> | 10 Name one thing that makes it misleading. | <input type="text"/> |

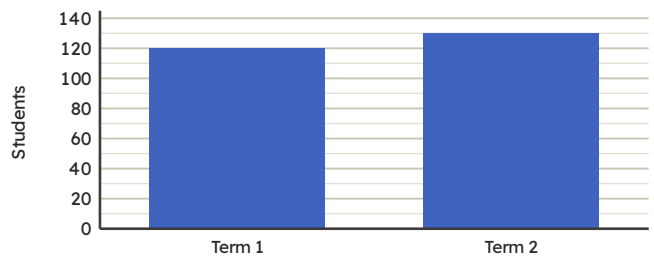
Level 2 · Building up

Graphs P and Q show the same data: 120 students bought lunch in Term 1 and 130 in Term 2.

Graph P: students buying lunch



Graph Q: students buying lunch



WE DO With your teacher: copy what they write on the board.

Example 3 Which graph makes the increase look bigger? Explain why.

Example 4 In Graph P, Term 2's bar looks 3 times as tall. Is that true? How many more students were there?

YOU DO On your own: Set C. Questions 1–5 use Graphs P and Q above.

1 Graph P: what does the vertical axis start at?

2 Graph Q: what does the vertical axis start at?

3 How many more students bought lunch in Term 2?

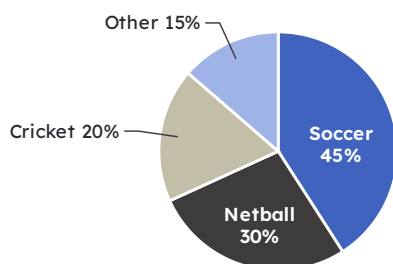
4 In Graph P, Term 2's bar looks how many times as tall as Term 1's?

5 Which graph would the canteen use to show a big jump? Why?

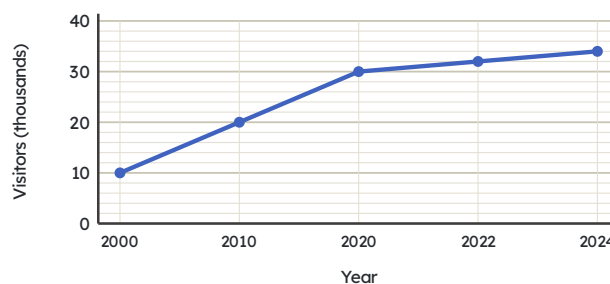
YOU DO

On your own: Set C continued. Questions 6–7 use the sector graph and 8–10 use the line graph.

Favourite sport of Year 7



Visitors to the town museum



6

Sector graph: add the percentages. What is the total?

9

Line graph: how many years are between 2020 and 2022?

7

Why is the sector graph misleading?

10

Why is the line graph's horizontal axis misleading?

8

Line graph: how many years are between 2000 and 2010?



Level 3 · Challenge

WE DO

With your teacher: copy what they write on the board.

Example 5

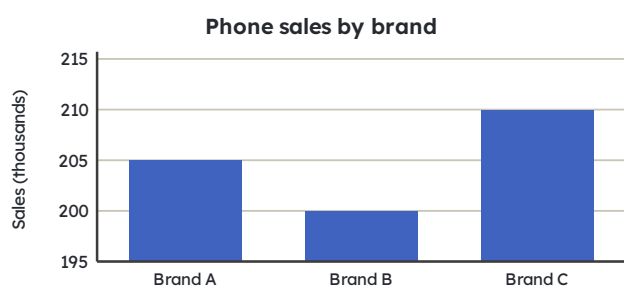
Redraw the class captain graph (page 11) fairly: Ava 52 votes, Ben 48 votes.

Example 6

Who might want to use the misleading class captain graph, and why?

YOU DO

On your own: Set D. Questions 1–3 use the phone sales graph.



- 1 What does the graph suggest about Brand C compared with Brand B?

- 2 What are the real sales of Brands B and C? What is the difference?

- 3 Redraw the phone sales graph fairly. Start the vertical axis at 0, add a title and label both axes.

YOU DO On your own: Set D continued.

- 4 Redraw the canteen sales graph (Set B) with the vertical axis starting at 0. Sales (\$ thousands) for 2019 to 2024: 41, 40, 43, 44, 45, 46.
- 5 How does your new canteen graph change the impression?
- 6 A headline says “Crime has doubled!” The graph’s axis starts at 50 and crimes rose from 52 to 54. Is the headline true? Explain.
- 7 A house is drawn twice as tall **and** twice as wide to show prices doubling. Why is this misleading?
- 8 Give two reasons someone might make a misleading graph on purpose.
- 9 List three things every fair graph should have.
- 10 A graph shows only the months when sales went up. Why is leaving data out misleading?

Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

A shop's sales doubled. On a poster, this year's circle has **twice the width and twice the height** of last year's. How many times bigger does it look?

1. Width $\times 2$ and height $\times 2$
2. The area (how big it looks) is $2 \times 2 = 4$ times as big
3. Sales only doubled, so the picture exaggerates



last year



this year

Answer: it looks 4 times bigger

YOU DO

On your own: read the worked example, then try these.

- E1** A picture is made 3 times as wide and 3 times as tall. How many times bigger does it look?

- E2** Sales went from 100 to 150. A bag is drawn 1.5 times as tall and 1.5 times as wide. How many times bigger does it look? How should it be drawn fairly?

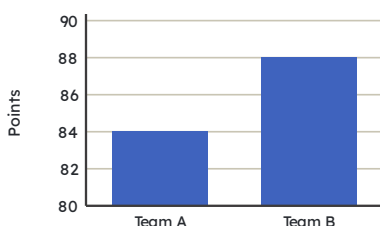
✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Points scored



Level 1

What number does the vertical axis start at?

Level 2

Team B's bar looks twice as tall. How many more points did B really score?

Level 3

Why is this graph misleading? How would you fix it?

Lesson summary

Missed the lesson? Start here.

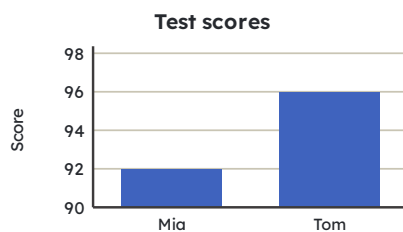
Is this graph fair?

1. Does the vertical axis start at 0 (or show a break)?
2. Is the **scale even** (equal gaps = equal amounts)?
3. Are the **title and axis labels** there?
4. Is **all the data** shown, with bars the same width?

Level 1

A column graph's vertical axis starts at 90. What is the problem?

1. Bar heights are measured from 90, not from 0
2. So small differences look huge

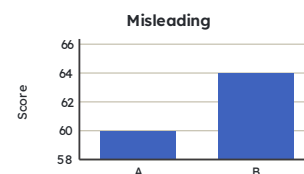


Answer: it is misleading. Start the axis at 0.

Level 2

Scores of 60 and 64 on an axis starting at 58. How many times as tall does the 64 bar look? What is the real difference?

1. Heights on the graph: $60 - 58 = 2$ and $64 - 58 = 6$
2. $6 \div 2 = 3$, so it looks 3 times as tall
3. Real difference: $64 - 60 = 4$

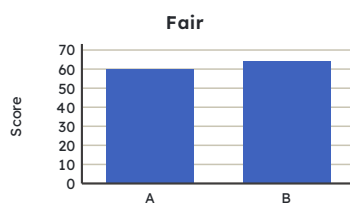


Answer: looks 3 times as tall; really only 4 more

Level 3

Redraw the Level 2 graph fairly.

1. Start the vertical axis at 0 with an even scale
2. Plot the same values: 60 and 64
3. Add a title and label both axes



Answer: the bars now look almost the same, which is the truth

Extension

A picture is drawn twice as wide and twice as tall. How many times bigger does it look?

1. Area grows by width $\times$ height
2. $2 \times 2 = 4$



Answer: 4 times bigger, though the value only doubled

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LESSON
10.03

Dot plots

Date _____

Learning intentions

We are learning to:

- read information from a dot plot
- draw a dot plot from a list of data
- describe a dot plot using the mode, clusters and outliers

Success criteria

I can:

- ☐ count how many times a value occurs and find the **mode**
- ☐ draw a neat dot plot with an even scale and a label
- ☐ spot an **outlier** and describe where the data **clusters**

Key terms Write the definition as your teacher explains it.

Dot plot

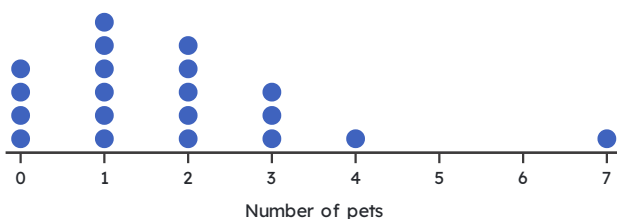
Outlier

Cluster

Notes

Level 1 · Getting started

Number of pets owned by students in 7B



WE DO

With your teacher: copy what they write on the board.

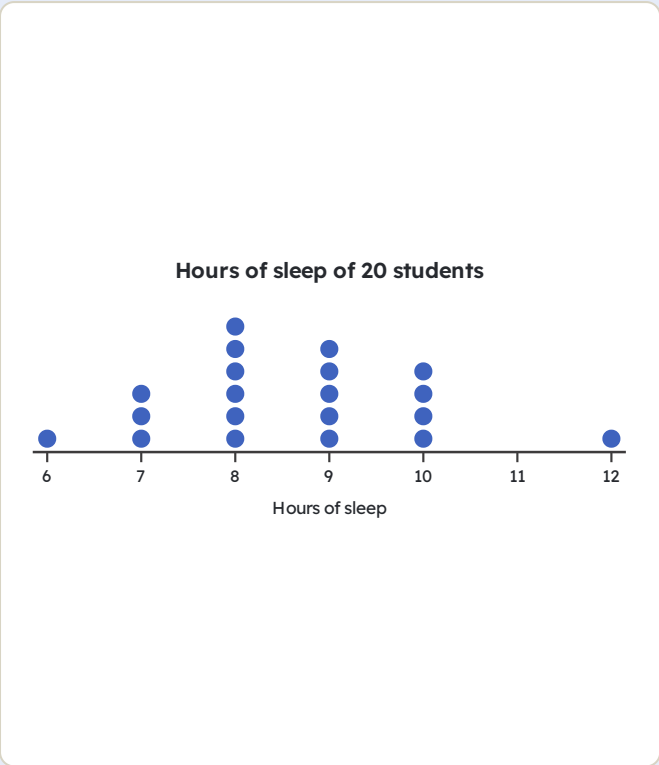
Example 1

How many students have 2 pets?
How many students were surveyed?

Example 2

What is the mode? Which value is an outlier?

YOU DO On your own: Set A. Use the sleep dot plot.



- 1 How many students were surveyed?
- 2 How many slept 8 hours?
- 3 How many slept 10 hours?
- 4 What is the mode?
- 5 What is the least sleep?
- 6 What is the most sleep?
- 7 What is the outlier?
- 8 How many slept 9 hours or more?
- 9 How many slept less than 8 hours?
- 10 How many more slept 8 hours than 7?

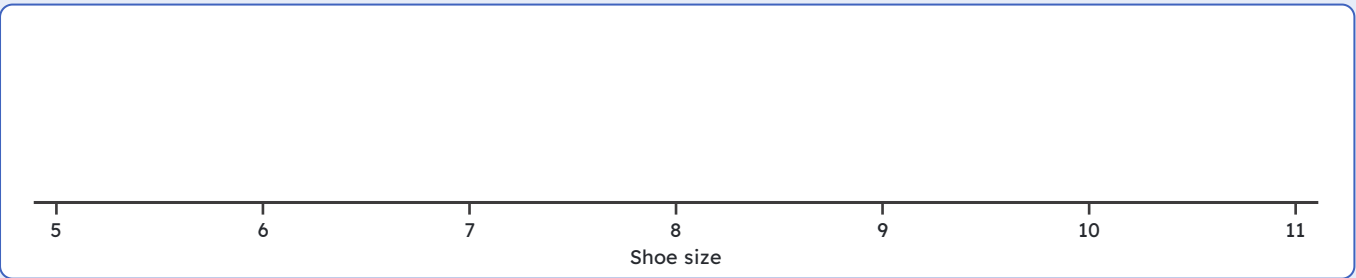
YOU DO On your own: Set B. Shoe sizes of 16 students.

Data: 6, 7, 7, 8, 8, 8, 9, 7, 8, 10, 9, 8, 7, 6, 8, 9

1 Complete the frequency table.

Size	6	7	8	9	10
Tally					
Frequency					

2 Draw a dot plot of the shoe sizes.

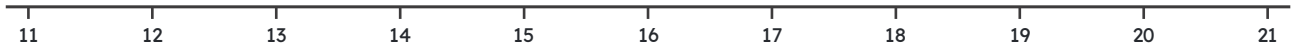


- 3 What is the mode?
- 4 How many wear size 9?
- 5 How many wear a size bigger than 8?
- 6 What is the smallest size?
- 7 How many wear size 7 or smaller?
- 8 Is there an outlier? (yes/no)
- 9 What fraction wear size 8?
- 10 Which size is least common?

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

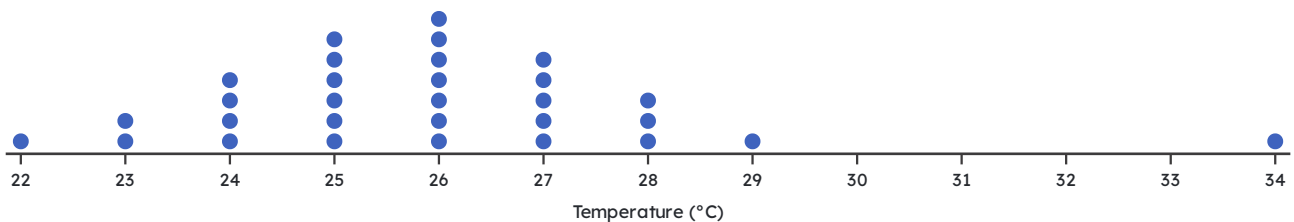
Example 3 Draw a dot plot for: 12, 14, 13, 12, 15, 12, 14, 20



Example 4 Describe the dot plot in Example 3. What is the mode? Where is the cluster? Is there an outlier?

YOU DO On your own: Set C. Use the temperature dot plot.

Daily maximum temperature, Kingscliff, November



1 What is the mode?

6 What percentage of days were 25 °C?

2 What is the outlier?

7 How many days were 24 or 25 °C?

3 How many days were 25 °C?

8 What fraction of days were 26 °C?

4 How many days were above 27 °C?

9 Where is the main cluster?

5 What fraction of days were below 24 °C?

10 Give a reason for the outlier.

**Level 3 · Challenge**

WE DO With your teacher: copy what they write on the board.

Test scores out of 10: 5, 6, 6, 7, 7, 7, 7, 8, 8, 8, 9, 9, 10, 2, 6, 7, 8, 9, 7, 8

Example 5 Draw a dot plot of the test scores. Choose your own scale and label it.



Write your own scale under the line, then label the axis.

Example 6 Describe the shape of the data. What does the dot plot tell the teacher about the class?

YOU DO

On your own: Set D. Text messages sent in one day by 25 students.

Data: 5, 7, 3, 6, 5, 12, 6, 4, 5, 8, 6, 7, 5, 30, 6, 5, 9, 7, 4, 6, 5, 8, 6, 7, 5

- 1 Draw a dot plot of the data. Choose a scale that fits every value, and label the axis.



Write your own scale under the line, then label the axis.

- 2 What is the mode?

- 4 How many students sent more than 7 texts?

- 3 Which value(s) are outliers?

- 5 What fraction sent 5 or fewer texts? Simplify.

YOU DO

On your own: Set D continued. Use your dot plot from the previous page.

- 6 What percentage sent exactly 6 texts?

- 7 Describe where the data clusters.

- 8 If the outlier is removed, does the mode change? Explain.

- 9 Why is a dot plot a good choice for this data?

- 10 Write one sentence that sums up what the dot plot shows.

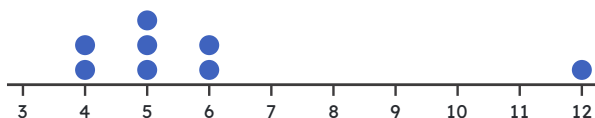
Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

Make up 8 values whose dot plot has a mode of 5, a cluster from 4 to 6, and an outlier at 12.

- Put the most dots at 5: 5, 5, 5
- Add values close by for the cluster: 4, 4, 6, 6
- Add one value far away for the outlier: 12



Answer: 4, 4, 5, 5, 5, 6, 6, 12

YOU DO

On your own: read the worked example, then try these.

- E1** Make up 10 values with a mode of 3 and one outlier. Draw the dot plot.



Write your own scale under the line, then label the axis.

- E2** Of 30 values, $\frac{1}{3}$ are 7s. There are twice as many 7s as 8s. How many 8s?

- E3** Why would a dot plot **not** suit the heights of 200 students?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

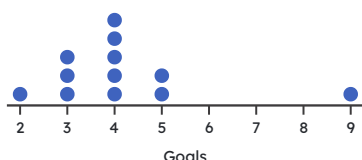
YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Goals in 12 netball games



Level 1

What is the mode?

Level 2

What fraction of the games had 3 goals or fewer? Simplify.

Level 3

Which value is the outlier? Where do the other values cluster?

Lesson summary

Missed the lesson? Start here.

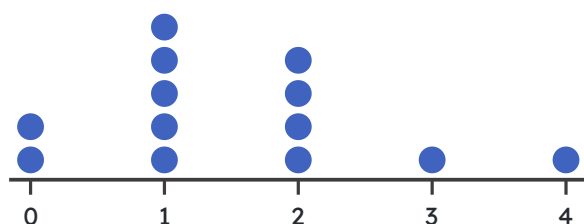
Drawing a dot plot

1. Find the smallest and largest values.
2. Draw a number line with an **even scale**.
3. Put one dot per value above the line. Stack neatly.
4. Add a **title** and label the axis.

Level 1

How many students have 2 siblings? What is the mode?

1. Count the dots above 2: there are 4
2. The tallest stack is above 1

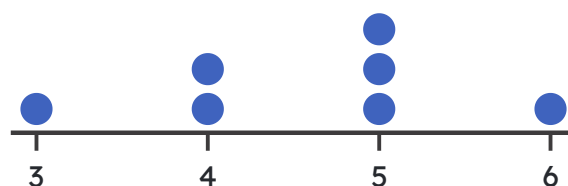


Answer: 4 students; mode 1

Level 2

Draw a dot plot for: 3, 5, 4, 5, 6, 5, 4

1. Smallest 3, largest 6: draw a line from 3 to 6
2. Put one dot above the line for each value

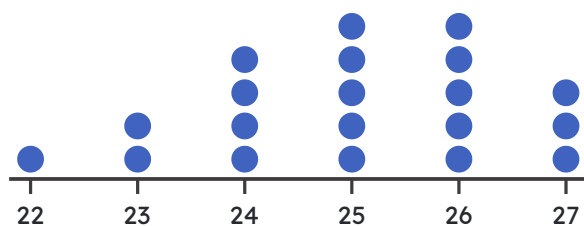


Answer: the dot plot below (mode 5)

Level 3

What fraction of these 20 days were warmer than 25 °C?

1. Count the dots above 26 and 27: $5 + 3 = 8$
2. Fraction = $8/20 = 2/5$

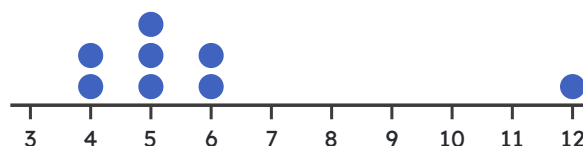


Answer: $2/5$ of the days (40%)

Extension

Describe the dot plot using a cluster and an outlier.

1. Most values bunch together from 4 to 6
2. 12 is far away from the rest



Answer: cluster from 4 to 6, outlier 12

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.04

Stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- read values from a stem-and-leaf plot
- put a stem-and-leaf plot in order
- draw a stem-and-leaf plot from a list of data

Success criteria

I can:

- ☐ split a number into its **stem** and **leaf** and read the key
- ☐ rewrite an unordered plot as an ordered plot
- ☐ draw an ordered stem-and-leaf plot with a key, including decimals

Key terms Write the definition as your teacher explains it.

Stem-and-leaf plot

Stem

Leaf

Notes

Level 1 · Getting started

Ages of people in a surf class

Stem	Leaf
1	2 5 8
2	0 3 3 7
3	1 4
4	6

Key: 2 | 3 = 23 years

WE DO

With your teacher: copy what they write on the board.

Example 1

Write all the ages in the 2 row.

Example 2

How many people are in the class?
Who is youngest and oldest?

YOU DO

On your own: Set A. Spelling test marks out of 50.

Stem	Leaf
2	4 8
3	0 2 5 5 9
4	1 3 3 3 6 8
5	0

Key: 3 | 5 = 35 marks

- 1 How many students sat the test?
- 2 What was the lowest mark?
- 3 What was the highest mark?
- 4 What is the mode?
- 5 How many scored in the 30s?
- 6 How many scored 40 or more?
- 7 What is the stem of 35?
- 8 What is the leaf of 48?
- 9 How many scored less than 30?
- 10 What number does 4 | 6 stand for?

YOU DO

On your own: Set B. Stems and leaves.

- 1 Stem and leaf of 47?
-

- 4 Stem and leaf of 9?
-

- 2 Stem and leaf of 63?
-

- 5 Order the leaves: 5 | 8 2
-
- 6 1
-

- 3 Stem and leaf of 105?
-

- 6 Order the leaves: 7 | 4 0
-
- 9 3 3
-

- 7 Draw an ordered stem-and-leaf plot of:
-
- 23, 31, 27, 35, 22, 40, 38, 31, 26, 44

Stem	Leaf
2	
3	
4	

Key: ____ | ____ = _____

- 8 How many values are in the 30s?
-

- 9 What is the mode?
-

- 10 What is the highest value?
-

Level 2 · Building up

WE DO With your teacher: copy what they write on the board.

Example 3 Rewrite this unordered plot as an ordered plot.

Unordered

Stem	Leaf
6	8 2 5
7	3 0 9 1
8	4 4 0

Key: 7 | 3 = 73

Stem	Leaf
6	
7	
8	

Key: ____ | ____ = _____

Example 4 Draw an ordered stem-and-leaf plot of: 56, 43, 61, 48, 55, 59, 62, 47, 55

Stem	Leaf

Key: ____ | ____ = _____

YOU DO On your own: Set C. Masses of rugby players (kg). Questions 3–10 are on the next page.

Unordered plot

Stem	Leaf
7	9 5 8 1 6
8	4 2 5 0 0 8 3
9	0 8 4 2 2 2
10	5 4

Key: 8 | 4 = 84 kg

1 Rewrite the plot as an ordered stem-and-leaf plot.

Stem	Leaf
7	
8	
9	
10	

Key: ____ | ____ = _____

2 How many players are there?

YOU DO

On your own: Set C continued. Use the rugby plot on the previous page.

3 What is the lowest mass?

7 How many weigh 90 kg or more?

4 What is the highest mass?

8 What fraction weigh in the 80s?

5 Which mass occurs most often?

9 What percentage weigh 100 kg or more?

6 How many players weigh less than 80 kg?

10 Why is an ordered plot easier to read?

Level 3 · Challenge

WE DO

With your teacher: copy what they write on the board.

Example 5

Heights (cm): 142, 155, 138, 149, 151, 160, 147, 138, 153. Use the first two digits as the stem.

Stem	Leaf

Key: ____ | ____ = _____

Example 6

Decimals: 2.4, 3.1, 2.8, 3.5, 4.0, 2.4, 3.9. Stem = whole number, leaf = tenths.

Stem	Leaf

Key: ____ | ____ = _____

YOU DO On your own: Set D. Decimals.

100 m sprint times (seconds): 14.2, 13.8, 15.1, 14.6, 13.5, 14.2, 16.0, 15.4, 14.9, 13.8, 14.2

- 1 Draw an ordered stem-and-leaf plot of the times. Include a key.

Stem	Leaf

Key: ____ | ____ = _____

- 4 What is the mode?

- 2 How many runners were there?

- 3 What was the fastest time?

- 6 What fraction ran 15 seconds or slower?

- 5 How many ran under 14 seconds?

- 7 What is the difference between the fastest and slowest times?

YOU DO On your own: Set D continued. 3-digit numbers.

Scores in a video game: 118, 124, 131, 109, 127, 124, 135, 112, 140

- 8 Draw an ordered stem-and-leaf plot of the scores. Include a key.

Stem	Leaf

Key: ____ | ____ = _____

- 9 Which stems did you use? Why?

- 10 Why must a stem-and-leaf plot have a key?

- 11 Give one advantage of a stem-and-leaf plot over a column graph.

Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example: split stems

Many values share a stem: 41, 42, 42, 44, 45, 46, 47, 47, 48, 49, 52, 53. Draw a split-stem plot.

- Split each stem into two rows
- Row "4" holds leaves 0 to 4; row "4*" holds leaves 5 to 9
- Do the same for 5 if you need to

Stem	Leaf
4	1 2 2 4
4*	5 6 7 7 8 9
5	2 3

Key: 4* | 7 = 47

Answer: the long row is split, so it is easier to read

YOU DO

On your own: read the worked example, then try these.

- E1** Draw a split-stem plot of: 20, 21, 23, 25, 26, 28, 31, 32, 37

Stem	Leaf
2	
2*	
3	
3*	

Key: ____ | ____ = ____

- E2** A plot has the key 5 | 2 = 0.52. What number does 6 | 7 stand for?

- E3** When is it useful to split the stems?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Unordered plot

Stem	Leaf
3	5 1 8
4	2 0 6 6
5	3

Key: 4 | 2 = 42

Level 1

How many values are in the plot?

Level 2

Write the 4 row with its leaves in order.

Level 3

What fraction of the values are 40 or more? What is the mode?

Lesson summary

Missed the lesson? Start here.

Drawing a stem-and-leaf plot

1. Choose the **stems** (usually the tens) and write them down the left.
2. Write each **leaf** (the units) next to its stem.
3. Rewrite with the leaves **in order**, smallest first.
4. Always add a **key**, e.g. $3 | 5 = 35$.

Level 1

What values are in the 3 row?

1. The stem 3 means 30-something
2. Put each leaf after the stem: 32, 35, 37

Stem	Leaf
2	4 8
3	2 5 7
4	1

Key: $3 | 2 = 32$

Answer: 32, 35 and 37

Level 2

Order this plot: $5 | 7 \ 2 \ 9 \ 6 | 3 \ 0 \ 8 \ 1$

1. Keep the stems
2. Sort each row's leaves from smallest to largest

Stem	Leaf
5	2 7 9
6	0 1 3 8

Key: $5 | 2 = 52$

Answer: $5 | 2 \ 7 \ 9$ and $6 | 0 \ 1 \ 3 \ 8$

Level 3

Draw a plot of 2.3, 3.6, 2.9, 3.1, 4.2

1. Stem = whole number, leaf = tenths
2. Stems 2, 3, 4
3. Order the leaves and add a key

Stem	Leaf
2	3 9
3	1 6
4	2

Key: $2 | 3 = 2.3$

Answer: key $2 | 3 = 2.3$

Extension

Why split a stem into 4 and 4*?

1. A row with lots of leaves is hard to read
2. 4 takes leaves 0–4 and 4* takes 5–9

Stem	Leaf
4	0 2 3
4*	5 7 8 9

Key: $4* | 7 = 4.7$

Answer: it spreads out crowded data

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.05

The mean and mode

Date _____

Learning intentions

We are learning to:

- calculate the mean and the mode of a set of data
- use the mean to find a missing value
- decide whether the mean or the mode better describes the data

Success criteria

I can:

- ☐ find the **mean** by adding the values and dividing by how many there are
- ☐ find the **mode**, and tell when there is no mode or more than one
- ☐ work backwards from the mean to find a missing value

Key terms Write the definition as your teacher explains it.

Mean

Mode

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Find the **mean** of: 4, 7, 5, 8, 6

Example 2 Find the **mode** of: 3, 9, 4, 7, 4, 8, 5

YOU DO

On your own: Set A. Find the mean of each set of numbers.

1 2, 4, 6**6** 6, 6, 9, 3**2** 5, 7, 9**7** 2, 8, 5, 7, 3**3** 1, 3, 5, 7**8** 12, 15, 18**4** 10, 20, 30**9** 4, 9, 2, 5, 10**5** 3, 5, 4, 8**10** 7, 11, 9, 13

YOU DO

On your own: Set B. Find the mode. Write “no mode” if there isn’t one.

1 2, 3, 3, 5, 8

6 4, 9, 7, 9, 2, 5, 9

2 1, 4, 4, 6, 9

7 1, 2, 3, 4, 5

3 7, 2, 5, 2, 9

8 6, 2, 8, 2, 6, 5

4 6, 1, 8, 6, 3

9 red, blue, red, green,
blue, red

5 10, 12, 11, 10, 15

10 20, 25, 22, 25, 21

Level 2 · Building up**WE DO**

With your teacher: copy what they write on the board.

Example 3

Find the mean and mode of: 5, 9, 8, 6, 9, 7. Round the mean to 2 decimal places.

Example 4

Find the mean and mode of the scores in this table.

Score	1	2	3	4
Frequency	2	5	4	1

YOU DO

On your own: Set C. Find the mean and the mode. Round the mean to 2 decimal places where needed.

- 1 5, 3, 2, 6, 7, 5, 5

- 6 Goals in 8 games: 2, 0, 3, 1, 2, 4, 2, 2

- 2 9, 5, 2, 2, 5, 2, 4, 7

- 7 Pocket money (\$): 15, 20, 10, 25, 20, 30

- 3 12, 15, 11, 15, 10, 14

- 8 Scores: 1 three times, 2 four times, 3 twice, 4 once

- 4 2.5, 3.5, 4, 6

- 9 Goals: 0 twice, 1 four times, 2 three times, 3 once

- 5 Temperatures ($^{\circ}\text{C}$) for a week: 21, 24, 19, 24, 22, 23, 21

- 10 Heights (cm): 150, 145, 160, 155, 148, 152



Level 3 · Challenge

WE DO

With your teacher: copy what they write on the board.

Example 5

The mean of five numbers is 8. Four of them are 6, 9, 5 and 10. Find the fifth number.

Example 6

Ages at a party: 11, 12, 12, 13, 12, 44. Find the mean and mode. Which better describes a typical age?

YOU DO

On your own: Set D. Show your working.

- 1 The mean of four numbers is 6. Three of them are 4, 7 and 8. Find the fourth.

- 3 Write a set of five numbers with a mean of 6 and a mode of 4.

- 2 The mean of five test scores is 72. Four are 68, 75, 70 and 80. Find the fifth.

- 4 For 3, 5, 6, 6, 40, find the mean and mode. Which is the better measure? Why?

YOU DO On your own: Set D continued. Show your working.

- 5 The number 20 is added to 4, 6, 7, 8, 10. How does the mean change?

- 8 Shoe sizes sold: 7, 8, 8, 9, 10, 8, 11. Should the shop use the mean or the mode to decide what to order? Why?

- 6 Mia scored 7, 8, 6 and 9 on four quizzes. What must she score next to have a mean of 8?

- 9 The mean of three whole numbers in a row (like 4, 5, 6) is 17. What are the numbers?

- 7 The mean of six numbers is 15. One is removed and the mean of the other five is 14. Which number was removed?

- 10 Class A: 12, 15, 15, 18, 20. Class B: 10, 14, 16, 19, 21. Compare their means and modes.

 Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

 Worked example

The mean age of four friends is 13. A fifth friend joins and the mean age becomes 14. How old is the new friend?

1. Total of four ages: $4 \times 13 = 52$
2. Total of five ages: $5 \times 14 = 70$
3. New friend: $70 - 52 = 18$
4. Check: $70 \div 5 = 14$ ✓

Key idea: mean $\times$ number of values = total.

Answer: 18 years old

YOU DO

On your own: read the worked example, then try these.

- E1** The mean mark of 10 students is 65. One test is re-marked and gains 15 marks. What is the new mean?

- E2** The mean of 8 numbers is 12. The numbers 5 and 9 are removed. What is the mean of the rest?

- E3** Can the mean be a number that is not in the data? Give an example.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

 Level 1

Find the mean of: 4, 6, 8, 10, 12

 Level 2

Find the mean and mode of: 3, 7, 7, 2, 9, 5

 Level 3

The mean of four numbers is 9. Three of them are 7, 12 and 5. Find the fourth number.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. **Mean** = total of the values $\div$ number of values.
 2. **Mode** = the value that appears most often.
 3. There can be **no mode** or **more than one** mode.
 4. An **outlier** pulls the mean towards it; the mode is not affected.

 Level 1

Find the mean and mode of 3, 5, 7, 5, 10

1. Total: $3 + 5 + 7 + 5 + 10 = 30$
2. 5 values, so $30 \div 5 = 6$
3. 5 appears most

Answer: mean 6, mode 5

 Level 2

Find the mean and mode of 7, 8, 6, 9, 9, 5, 8

1. Total = 52, 7 values
2. $52 \div 7 = 7.428... \approx 7.43$
3. 8 and 9 both appear twice

Answer: mean 7.43, modes 8 and 9

 Level 3

The mean of 4 scores is 15. Three are 12, 18, 14. Find the fourth.

1. Total needed: $4 \times 15 = 60$
2. Known total: 44
3. $60 - 44 = 16$

Answer: 16

 Extension

5 numbers have a mean of 10. A sixth is added and the mean is 12. What was added?

1. $5 \times 10 = 50$
2. $6 \times 12 = 72$
3. $72 - 50 = 22$

Answer: 22

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.06

The median and range

Date _____

Learning intentions

We are learning to:

- find the median and range of a set of data
- find the median when there are two middle values
- use the median and range to compare and describe data

Success criteria

I can:

- ☐ put data **in order** and find the middle value
- ☐ find the median of an even number of values
- ☐ find the **range** by subtracting the lowest value from the highest

Key terms Write the definition as your teacher explains it.

Median

Range

Notes

Level 1 · Getting started

WE DO With your teacher: copy what they write on the board.

Example 1 Find the **median** of: 3, 9, 4, 7, 4, 8, 5

Example 2 Find the **range** of: 12, 5, 9, 20, 7

YOU DO

On your own: Set A. Find the median. Put the numbers in order first!

1 2, 3, 3, 5, 8**6** 4, 9, 7, 9, 2, 5, 9**2** 1, 4, 4, 6, 9**7** 15, 13, 14, 13, 18**3** 7, 2, 5, 2, 9**8** 8, 3, 6, 3, 1, 5, 3**4** 6, 1, 8, 6, 3**9** 20, 25, 22, 25, 21**5** 10, 12, 11, 10, 15**10** 5, 7, 6, 7, 9, 4, 8

YOU DO On your own: Set B. Find the range (highest – lowest).

1 3, 8, 5, 10

6 6, 6, 6, 6

2 14, 6, 23, 9, 11

7 45, 12, 33, 50, 19

3 1, 1, 6, 4

8 7.5, 3.2, 6.1

4 25, 30, 18, 22

9 0, 15, 9, 3

5 100, 85, 92, 97

10 -2, 5, 1, 3

Level 2 · Building up**WE DO** With your teacher: copy what they write on the board.**Example 3** Find the median and range of: 8, 3, 10, 5, 7, 12**Example 4** Find the median and range of the data in this stem-and-leaf plot.

Stem	Leaf
2	4 8
3	1 5 5 9
4	2

Key: 3 | 1 = 31

YOU DO

On your own: Set C. Find the median and the range.

1 4, 9, 2, 7**2** 11, 15, 12, 18, 14, 20**3** 3.5, 1.2, 4.8, 2.6, 3.1**4** 9, 2, 6, 11, 4, 8**5** Temperatures ($^{\circ}\text{C}$): -4, 2, -1, 5, 0, 3**6** Temperatures ($^{\circ}\text{C}$) for a week: 21, 24, 19, 24, 22, 23, 21**7** Scores: 1 three times, 2 four times, 3 twice, 4 once**8** Heights (cm): 150, 145, 160, 155, 148, 152**9** Prices (\$): 100, 250, 175, 300, 125**10** Goals: 0 twice, 1 four times, 2 three times, 3 once

 Level 3 · Challenge**WE DO** With your teacher: copy what they write on the board.**Example 5** Ages at a party: 11, 12, 12, 13, 12, 44.
Find the median, range and mean.
Which describes a typical age best?**Example 6** Write five numbers with a median of 8
and a range of 10. The smallest is 3.**YOU DO** On your own: Set D. Show your working.

- 1 Write a set of five numbers with a median of 6 and a range of 8.

- 2 The range of a data set is 15 and the smallest value is 7. What is the largest value?

- 3 The number 20 is added to 4, 6, 7, 8, 10. How do the median and range change?

- 4 Class A: median 65, range 40. Class B: median 60, range 10. Which class is more consistent? Which did better?

YOU DO On your own: Set D continued. Show your working.

- 5 Explain why one very large value changes the mean a lot but the median only a little.

- 8 Heights (cm): 180, 175, 190, 182 and one more. The median is 182 and the range is 20. Find the missing height.

- 6 Six numbers have a median of 10. Four are 4, 8, 13 and 15. The other two are equal. What are they?

- 9 Can the range of a data set be 0? Explain.

- 7 Write five numbers with median 8, range 10, smallest 3 and mean 8.

- 10 Ages in a family: 42, 40, 14, 11, 8, 1. Find the median and range. Does the median describe this family well?

Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

25 values are written in order. Which value is the median?

1. Add 1 to the number of values: $25 + 1 = 26$
2. Halve it: $26 \div 2 = 13$
3. The median is the 13th value (12 values sit on each side)

Key idea: the median is in position $(n + 1) \div 2$. If that ends in .5, average the two values either side.



Answer: the 13th value

YOU DO

On your own: read the worked example, then try these.

- E1** There are 40 values in order. Which two positions are used to find the median?

- E2** Scores: 1 three times, 2 five times, 3 eight times, 4 four times. Find the median.

- E3** Find the median of all the whole numbers from 1 to 100.

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Level 1

Find the median of: 9, 3, 7, 5, 11

Level 2

Find the median and range of: 9, 3, 7, 3, 11, 6

Level 3

Write five numbers with a median of 6 and a range of 10. The smallest is 2.

Lesson summary

Missed the lesson? Start here.

- Remember**
1. Put the data **in order** first.
 2. **Median** = the middle value.
 3. Two middle values? Add them and **divide by 2**.
 4. **Range** = highest – lowest.

Level 1

Find the median of 7, 3, 9, 5, 4

1. Order: 3, 4, 5, 7, 9
2. Middle (3rd) value: 5

Answer: 5**Level 2**

Find the median and range of 6, 2, 9, 4

1. Order: 2, 4, 6, 9
2. Two middle values: $(4 + 6) \div 2 = 5$
3. Range: $9 - 2 = 7$

Answer: median 5, range 7**Level 3**Temperatures ranged from -3°C to 8°C . Find the range.

1. Range = highest – lowest
2. $8 - (-3) = 8 + 3 = 11$

Answer: 11°C **Extension**

Where is the median of 31 ordered values?

1. $(31 + 1) \div 2 = 16$

Answer: the 16th value

This space is left blank on purpose. It is the back of your exit ticket.

LESSON
10.07

Analysing dot plots and stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- find the range, median, mode and mean from a dot plot
- find the same statistics from a stem-and-leaf plot
- choose the best measure for data with an outlier

Success criteria

I can:

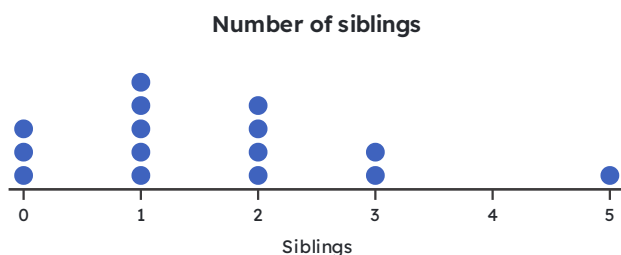
- ☐ use the **frequency** of each value to find the total and the mean
- ☐ count to the middle position of a plot to find the **median**
- ☐ explain how an **outlier** changes the mean and median

Key terms Write the definition as your teacher explains it.

Frequency
Measure of centre
Spread

Notes

Level 1 · Getting started


WE DO

With your teacher: copy what they write on the board.

Example 1

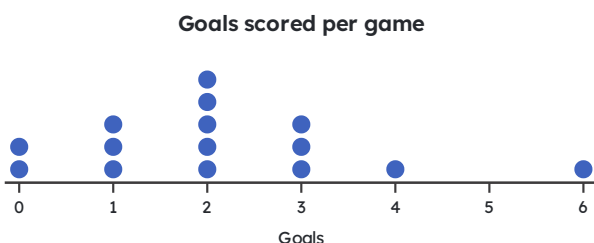
Find the range and the mode.

Example 2

Find the median and the mean.

YOU DO

On your own: Set A. Use the goals dot plot.



- 1 How many games were played?
- 2 What is the most goals in a game?
- 3 What is the range?
- 4 What is the mode?
- 5 What is the median?
- 6 How many goals were scored in total?
- 7 What is the mean (2 decimal places)?
- 8 Which value is an outlier?
- 9 In how many games were more than 2 goals scored?
- 10 What fraction of games had 0 goals?

YOU DO

On your own: Set B. Ages of people in a yoga class.

Stem	Leaf
2	4 6
3	1 1 5 8
4	0 2 7
5	3

Key: 3 | 1 = 31 years

- 1 How many people are in the class?
- 2 Who is the youngest?
- 3 Who is the oldest?
- 4 What is the range?
- 5 What is the mode?
- 6 What is the median?
- 7 What is the total of the ages?
- 8 What is the mean age?
- 9 How many are under 30?
- 10 What does 3 | 5 stand for?



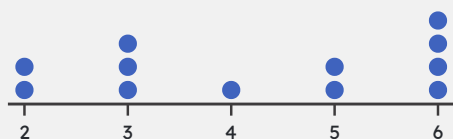
Level 2 · Building up

WE DO

With your teacher: copy what they write on the board.

Example 3

Find the range, median, mode and mean.

**Example 4**

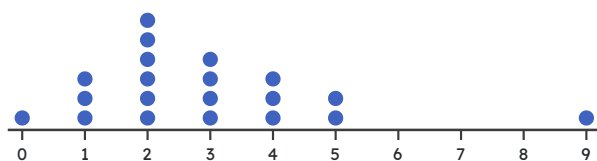
Find the range, median, mode and mean.

Stem	Leaf
1	5 8
2	0 3 3 7
3	1 4 4 4 9
4	2

Key: 2 | 3 = 23

YOU DO

On your own: Set C. Questions 1–5 use the books dot plot and 6–10 use the homework plot.

Books read in a month**1** Books: find the range.**2** Books: find the mode.**3** Books: find the median.**4** Books: find the mean.**5** Books: which value is an outlier? Does it make the mean bigger or smaller?**Minutes spent on homework**

Stem	Leaf
1	0 5 5 8
2	0 2 5 5 5 9
3	0 4 5
4	5

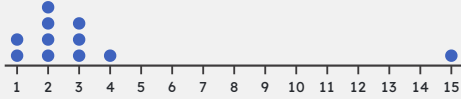
Key: 2 | 5 = 25 min

6 Homework: find the range.**7** Homework: find the mode.**8** Homework: find the median.**9** Homework: find the mean (2 d.p.).**10** Homework: how many students spent more than 30 minutes?

Level 3 · Challenge

WE DO With your teacher: copy what they write on the board.

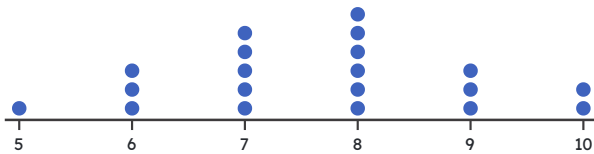
Example 5 Find the mean and median. Then remove the outlier and find them again.



Example 6 Which measure (mean, median or mode) best describes the data in Example 5? Explain.

YOU DO On your own: Set D. Questions 1–4 use the test scores dot plot.

Test scores out of 10



1 Find the mean, median, mode and range of the test scores.

2 A student who was away scores 2. Find the new mean and median.

3 Which measure changed more in Question 2? Why?

4 Could one more student raise the class mean to 8? Explain.

YOU DO

On your own: Set D continued. Questions 5–8 use the concert plot.

Ages at a concert	
Stem	Leaf
1	5 6 7 7 8 9
2	0 1 1 1 3 5
3	2 4
4	
5	
6	8

Key: 2 | 1 = 21 years

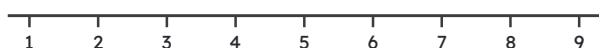
- 5 Find the mean, median, mode and range of the concert ages.

- 6 Which age is an outlier? How can you tell?

- 8 Which measure best describes a typical age at the concert? Why?

- 7 Remove the outlier. Find the new mean and range.

- 9 Draw a dot plot of 9 values with mode 4, median 5, range 6 and smallest value 2.



- 10 Why are the mean and median of a symmetric dot plot about the same?

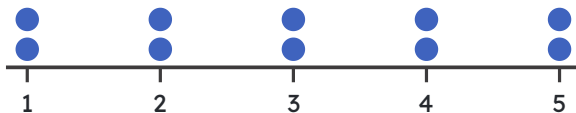
Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

A dot plot shows 10 values with a mean of 3. Nine of the values are 1, 2, 2, 3, 3, 4, 4, 5, 5. What is the tenth value?

1. Total of all 10 values: $10 \times 3 = 30$
2. Total of the nine values: 29
3. Tenth value: $30 - 29 = 1$



Answer: 1

YOU DO

On your own: read the worked example, then try these.

- E1** A stem-and-leaf plot shows 23, 25, 2?, 30, 34. The mean is 28. What is the missing leaf?

- E2** Add one value to 4, 5, 5, 6, 9 so that the median becomes 5.5. Which values work?

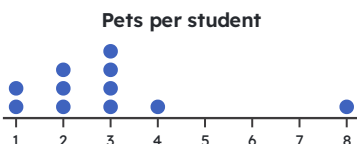
- E3** Make up six values where the mean is bigger than the median. Explain how you did it.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

What is the mode?

Level 2

Find the median and the range.

Level 3

Find the mean. What happens to the mean if the 8 is removed?

Lesson summary

Missed the lesson? Start here.

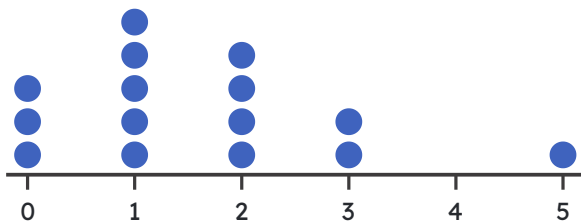
- From a plot**
- 1. Range:** highest – lowest.
 - 3. Median:** count to the middle position.

- 2. Mode:** tallest stack or most repeated leaf.
- 4. Mean:** add every value (value × frequency), then divide.

Level 1

Find the range and mode of the siblings dot plot.

- 1. Range:** highest – lowest = $5 - 0 = 5$
- 2. The tallest stack is above 1**



Answer: range 5, mode 1

Level 2

Find the median of this plot.

- 1. There are 7 values, so the median is the 4th**
- 2. Count along: 21, 24, 24, 30**

Stem	Leaf
2	1 4 4
3	0 5
4	2 6

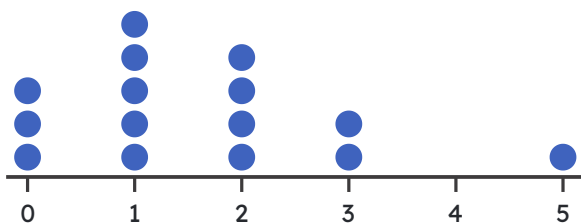
Key: 2 | 1 = 21

Answer: 30

Level 3

Find the mean of the siblings dot plot.

- 1. Total:** $0 \times 3 + 1 \times 5 + 2 \times 4 + 3 \times 2 + 5 \times 1 = 24$
- 2. There are 15 values:** $24 \div 15 = 1.6$

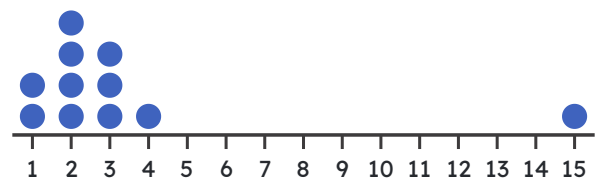


Answer: 1.6

Extension

Why does an outlier change the mean more than the median?

- 1. The mean adds up every value, so one big value adds a lot**
- 2. The median only uses the middle position**



Answer: the outlier pulls the mean towards it

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LESSON 10.08

Comparing data sets

Date _____

Learning intentions

We are learning to:

- compare two data sets using measures of centre and spread
- draw and read a back-to-back stem-and-leaf plot
- tell the difference between primary and secondary data

Success criteria

I can:

- ☐ compare the **mean** or **median** to decide which group did better
- ☐ compare the **range** to decide which group is more consistent
- ☐ draw a **back-to-back stem-and-leaf plot** and write a conclusion

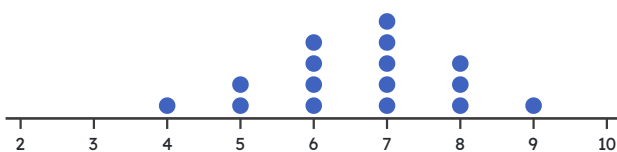
Key terms Write the definition as your teacher explains it.

Back-to-back stem-and-leaf plot
Primary data
Secondary data

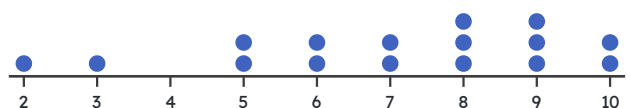
Notes

Level 1 · Getting started

Class A quiz scores



Class B quiz scores



WE DO

With your teacher: copy what they write on the board.

Example 1

Find the mode and the range of each class.

Example 2

Find the median of each class. Which class did better? Which was more consistent?

YOU DO On your own: Set A. Ages of people at two gyms.

Gym A	Stem	Gym B
8	1	
7 5 5 2	2	1 4 9
8 4 3 1	3	3 5 5 6 9
5 2	4	1 4 6 8
1	5	2 7
	6	3

Key: 3 | 5 = 35 years

- 1 How many people go to Gym A?
- 2 How many people go to Gym B?
- 3 Who is the youngest at Gym A?
- 4 Who is the oldest at Gym B?
- 5 What is the mode for Gym A?
- 6 What is the mode for Gym B?
- 7 What is the range for Gym A?
- 8 What is the range for Gym B?
- 9 How many people are in their 30s altogether?
- 10 Which gym has older people in general?

YOU DO On your own: Set B. Is each one primary data (you collect it yourself) or secondary data (someone else collected it)? Write P or S.

- 1 You measure the heights of your class.

- 6 You copy a population table from a textbook.

- 2 You download rainfall data from the Bureau of Meteorology.

- 7 You time your friends running 100 m.

- 3 You hand out a survey at lunch.

- 8 You use sports results from a news website.

- 4 You use Census results from the ABS.

- 9 You toss a coin 50 times and record the results.

- 5 You count cars driving past the school gate.

- 10 You use data from a science journal.

Level 2 · Building up

WE DO

With your teacher: copy what they write on the board.

Spelling scores. Class P: 34, 41, 45, 28, 39, 47, 52, 36 Class Q: 25, 31, 38, 29, 42, 33, 27, 35

Example 3

Draw a back-to-back stem-and-leaf plot of the scores.

Class P	Stem	Class Q
	2	
	3	
	4	
	5	

Key: ____ | ____ = _____

Example 4

Find the median and range of each class. Compare the two classes.

YOU DO

On your own: Set C. Heights (cm) of two Under-14 basketball squads.

Girls: 163, 157, 170, 158, 152, 166, 160, 148, 161, 161, 155**Boys:** 156, 150, 145, 158, 158, 158, 171, 153, 167, 149, 162, 175

- 1 Draw a back-to-back stem-and-leaf plot. Put the girls on the left.

Girls	Stem	Boys
	14	
	15	
	16	
	17	

Key: ____ | ____ = _____

- 2 Who is the tallest player? Which squad are they in?

- 5 Find the range of each squad.

- 6 Find the mean of each squad (2 decimal places).

- 7 Which squad is taller on average? Use a statistic.

- 8 Which squad has heights that are more alike? Use a statistic.

- 3 Find the median height of each squad.

- 4 Find the mode of each squad.

- 9 How many players in each squad are 160 cm or taller?

- 10 Write two sentences comparing the squads.

- 11 The coach measured the players herself. Is this primary or secondary data?

- 12 A coach wants the taller squad for a tournament. Which squad would you pick? Use two statistics.

 Level 3 · Challenge

WE DO With your teacher: copy what they write on the board.

Goals in 6 netball games. Amy: 12, 14, 13, 15, 14, 12 Beth: 5, 22, 8, 20, 18, 11

Example 5 Find the mean and the range for each player. Who is more consistent?

Example 6 Which player would you pick for the grand final? Use the statistics to justify your choice.

YOU DO On your own: Set D. Assignment marks (out of 100) for two classes.

7R: 52, 78, 72, 72, 86, 67, 61, 68, 81, 75, 63, 70, 64, 45, 58

7T: 66, 79, 92, 60, 74, 55, 23, 49, 88, 95, 83, 38, 90, 71, 97

- 1 Draw a back-to-back stem-and-leaf plot. Put 7R on the left.

7R	Stem	7T
	2	
	3	
	4	
	5	
	6	
	7	
	8	
	9	

Key: ____ | ____ = ____

- 2 What was the highest mark? Which class scored it?

- 3 Find the mean, median, mode and range of 7R.

- 4 Find the mean, median, mode and range of 7T.

YOU DO**On your own: Set D continued. Use your plot and statistics from the previous page.**

- 5 Which class did better overall? Use the mean or median to explain.

- 8 Remove that mark. Find the new mean and range for 7T.

- 6 Which class was more consistent? Use the range to explain.

- 9 The teacher got the marks from her own mark book. Is this primary or secondary data? Explain.

- 7 Which mark in 7T could be called an outlier? Why?

- 10 Write a conclusion comparing 7R and 7T using at least two statistics.



Extension · Stretch yourself

Finished Level 3? Stretch yourself here.



Worked example: combining groups

Class A has 10 students with a mean of 70. Class B has 15 students with a mean of 80. What is the mean of all 25 students?

1. Total for A: $10 \times 70 = 700$
2. Total for B: $15 \times 80 = 1200$
3. Total for everyone: $700 + 1200 = 1900$
4. Mean: $1900 \div 25 = 76$

Answer: 76 (not 75, because Class B is bigger)

YOU DO

On your own: read the worked example, then try these.

- E1** A group of 8 has a mean of 12 and a group of 12 has a mean of 17. Find the mean of all 20.

- E2** Why is the combined mean not always halfway between the two means?

- E3** Make up two sets of 5 values with the same mean but different ranges.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO

Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____

Team A	Stem	Team B
5 2	1	8
4 4 1	2	2 5 7 8 9
0	3	

Key: 2 | 4 = 24

Level 1

What is the highest score? Which team scored it?

Level 2

Find the median of each team.

Level 3

Which team was more consistent? Explain using the range.

Lesson summary

Missed the lesson? Start here.

Comparing data sets

- Centre** (mean or median): which group is higher?
- Spread** (range): smaller range = more consistent.
- Look for **outliers** that change the mean.
- Primary** data you collect; **secondary** data someone else collected.

Level 1

In this back-to-back plot, what does the left leaf 3 on stem 2 stand for?

- Left leaves belong to the left group (Team A)
- Stem 2 and leaf 3 make 23

Team A	Stem	Team B
7 3	2	1 5
4	3	0 2 6

Key: 2 | 3 = 23

Answer: a score of 23 for Team A

Level 2

Medians: Class X 64, Class Y 71. Ranges: X 20, Y 45. Compare the classes.

- Y has the higher median, so Y did better on average
- X has the smaller range, so X is more consistent

Answer: Y scored higher; X was more consistent

Level 3

Amy: mean 13.3, range 3. Beth: mean 14, range 17. Who is more reliable?

- The means are close
- Amy's range is much smaller, so her scores are more alike

Answer: Amy is more reliable

Extension

4 values have a mean of 5 and 6 values have a mean of 10. What is the combined mean?

- Totals: $4 \times 5 = 20$ and $6 \times 10 = 60$
- Combined: $80 \div 10 = 8$

Answer: 8 (not 7.5, because the second group is bigger)

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